RAKSHA R

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Professional Summary

Computer Science & Engineering undergraduate and Smart India Hackathon (SIH) 2025 National Winner with project and internship experience in software development, object-oriented programming, backend development, cloud computing, and machine learning. Develops applications using Java, Python, JavaScript, SQL, React, Spring Boot, Flask, and REST APIs. Familiar with Git/GitHub, CI/CD workflows, database concepts, debugging, troubleshooting, and collaborative software development.

Technical Skills

Programming: Java, Python, JavaScript, SQL, Object-Oriented Programming (OOP)
Web & Backend: React, Next.js, Node.js, Spring Boot, Flask, REST APIs, Application Development
Databases & Tools: MySQL, SQLite, MongoDB, SQL Syntax, Postman, Git, GitHub, Maven
Cloud & DevOps: AWS (API Gateway, IAM, S3), Azure IoT, Jenkins, CI/CD, Linux, Software Deployment
Engineering Practices: Software Development, Software Engineering Principles, Clean and Efficient Code, Debugging, Code Quality, Problem-Solving, Analytical Skills, Team Collaboration, Technical Communication, Agile Development

Experience

Software & Cloud Computing Intern *Jan 2024 – Apr2024*
Tru Projects | Hyderabad, India

- Developed AWS cloud data pipelines for a 5G Smart Diabetes healthcare analytics project, preparing healthcare datasets for machine learning analysis.
- Assisted with deployment, monitoring, and management of machine learning models across cloud platforms to support reliable operation.

AI & Machine Learning Intern *Jun 2025 – Jul2025*
Robomanthan (with BNM Institute of Technology) | Bengaluru, India

- Completed practical assignments and workshops in artificial intelligence, machine learning, and robotics.
- Worked with sensors, microcontrollers, and automation concepts for robotics and embedded-AI applications.

Projects

Digital Contact Tracing & Screening Tool for MDR Pathogens – SIH 2025 National Winner ***Flutter, XGBoost, AWS, ONNX Runtime***

- Developed an offline-first, cross-platform hospital contact-tracing application using on-device inference across 19 clinical features and a hybrid risk score combining machine learning (70%) with rules (30%).

WhatsApp Digital Forensics Investigation Tool ***AES-256-GCM, SQLite, SHA-256***

- Built a forensic analysis tool for WhatsApp .crypt14 database files, including chain-of-custody hashing and an interactive evidence dashboard.

DevOps Video-to-Audio Converter ***Spring Boot, Maven, FFmpeg, CI/CD***

- Implemented a Spring Boot backend using a Plan–Code–Build–Test–Release–Deploy workflow in a Maven-managed, CI/CD-ready architecture.

Cotton Leaf Disease Detection – ACICS 2025 Research Presenter ***CNN, TensorFlow/Keras, OpenCV***

- Trained a CNN for four-class cotton leaf disease detection, achieving 94.2% validation accuracy; presented the work at the ACICS 2025 international conference.

Education

B.E. in Computer Science & Engineering Aug 2024 – Jan 2027
BNM Institute of Technology, Bengaluru | CGPA: 8.85/10

Diploma in Computer Science & Engineering Jan 2021 – Jun 2024
Government Polytechnic College | CGPA: 9.6/10

Certifications

Azure IoT Fundamentals (Microsoft); NPTEL – Introduction to IoT; RPA Basics & Introduction to UiPath; Cisco Network Addressing

Achievements

- Smart India Hackathon (SIH) 2025 National Winner for the Digital Contact Tracing and Screening Tool for MDR Pathogens in Hospitals.
- Research Presenter, ACICS 2025 International Conference, presenting cotton leaf disease detection using CNN (94.2% validation accuracy).